

HARWELL RSX CHARACTERIZATION REPORT

OVERVIEW

Each resistor undergoes characterization at multiple temperature points to develop a precise model for calculating resistance across its operating range. Our modern resistors use a proprietary metal foil design that, depending on the unique assembly of each resistor, may be characterized via a polynomial or third-degree fit.

INSTRUMENT INFORMATION

Date: 13 NOV 2025  
Manufacturer: Harwell  
Model: RS-3  
Serial Number: RS30054  
Nominal Resistance: 10,000 Ω  
Calibrated Resistance: 9999.93600 Ω

TCR MODEL PARAMETERS

Calibrated Resistance: 9999.93600 Ω (at 23°C)

Straight-Line TCR: 0.013 ppm/°C  
Model Slope (at ref): 0.023 ppm/°C  
  
Alpha (α): 2.325E-08 Ω/Ω/°C  
Beta (β): 5.145E-09 Ω/Ω/°C²

RESISTANCE CALCULATION MODEL

To calculate the resistor value at a temperature other than the reference temperature (23.1°C), use this quadratic model:

$$R(T) = R_{(23)} + \alpha(T - 23) + \beta(T - 23)^2$$

Where:  
T = Desired temperature (°C)  
R<sub>(23)</sub> = 9999.93600 Ω  
α = 2.325E-04 Ω/°C  
β = 5.145E-05 Ω/°C²

⚠ Note: This model may not be accurate outside the characterized temperature range.

RESISTANCE vs TEMPERATURE TABLE

| Temperature (°C) | Temperature (°F) | Resistance (Ω) | Deviation (ppm) |
|------------------|------------------|----------------|-----------------|
| 18.0             | 64.4             | 9999.936124    | 0.01            |
| 18.5             | 65.3             | 9999.935996    | -0.00           |
| 19.0             | 66.2             | 9999.935893    | -0.01           |
| 19.5             | 67.1             | 9999.935817    | -0.02           |
| 20.0             | 68.0             | 9999.935766    | -0.02           |
| 20.5             | 68.9             | 9999.935740    | -0.03           |
| 21.0             | 69.8             | 9999.935741    | -0.03           |
| 21.5             | 70.7             | 9999.935767    | -0.02           |
| 22.0             | 71.6             | 9999.935819    | -0.02           |
| 22.5             | 72.5             | 9999.935897    | -0.01           |
| 23.0             | 73.4             | 9999.936000    | 0.00            |
| 23.5             | 74.3             | 9999.936129    | 0.01            |
| 24.0             | 75.2             | 9999.936284    | 0.03            |
| 24.5             | 76.1             | 9999.936465    | 0.05            |
| 25.0             | 77.0             | 9999.936671    | 0.07            |
| 25.5             | 77.9             | 9999.936903    | 0.09            |
| 26.0             | 78.8             | 9999.937161    | 0.12            |
| 26.5             | 79.7             | 9999.937444    | 0.14            |
| 27.0             | 80.6             | 9999.937753    | 0.18            |
| 27.5             | 81.5             | 9999.938088    | 0.21            |
| 28.0             | 82.4             | 9999.938449    | 0.24            |

# Temperature Coefficient of Resistance (TCR)

